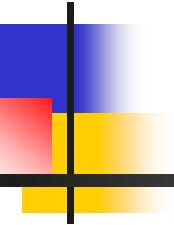


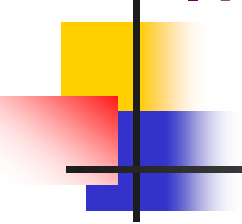
Deterioration of Eyesight in a Diabetic Patient



Endocrinology
2025

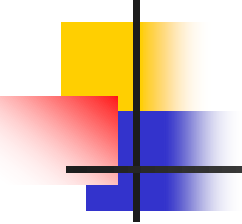
Prof Karen Lam (Medicine)

Prof Christopher Leung (Ophthalmology)



Important ocular complications of diabetes mellitus

- Diabetic retinopathy (non-proliferative vs. proliferative)
- Diabetic macular edema
- Neovascular glaucoma (growth of new vessels on iris extending to the angle of the anterior chamber, leading to high intraocular pressure and optic nerve degeneration)
- III, IV, VI nerve palsies
- Cataract



Diabetic retinopathy (DR)

Non-Proliferative

Microaneurysms

Dot & blot haemorrhages

Exudates: soft, hard

Proliferative

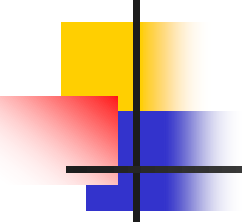
Retinal neovascularization

Vitreous haemorrhage

Tractional retinal detachment

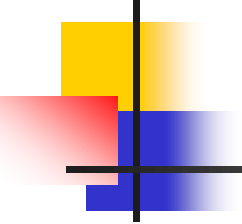
Soft exudates – swollen nerve fiber (whitish fluffy lesions)

Hard exudates – lipid and proteinaceous materials from chronic vascular leakage (yellowish discrete lesions)



Complications

- Abnormal angiogenesis leading to:
preretinal hemorrhage, vitreous
hemorrhage, and neovascular glaucoma.
- Abnormal fibrous proliferation leading to
tractional retinal detachment



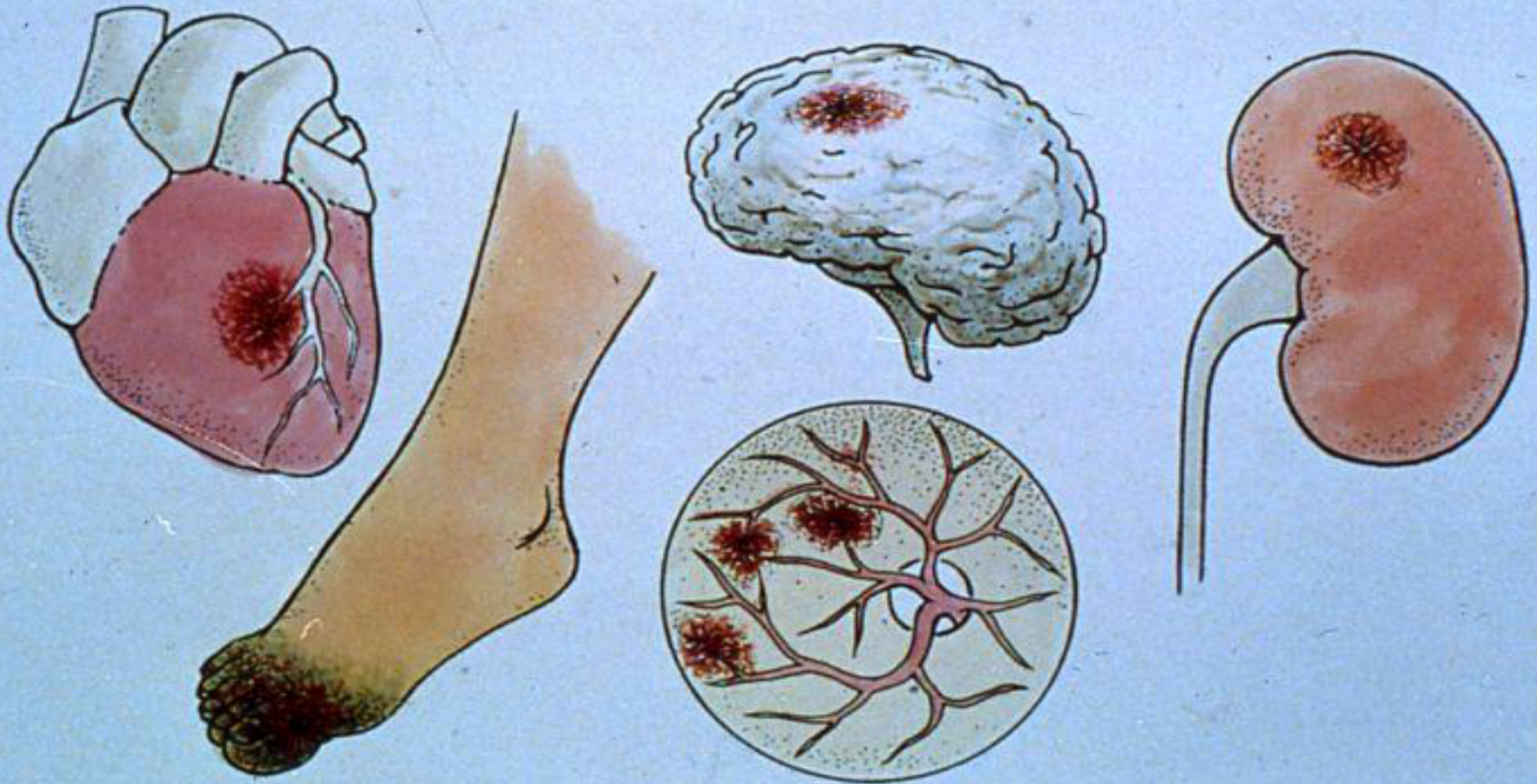
Treatment

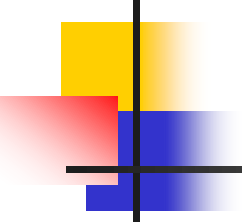
PREVENTION IS ALWAYS THE BEST CURE!!!

1. Proliferative diabetic retinopathy:
 - New blood vessels: panretinal photocoagulation;
anti-VEGF (vascular endothelial growth factor) monoclonal antibody
 - Vitreous haemorrhage/Retinal detachment: vitrectomy and laser
2. Clinically significant macular oedema: Focal or grid laser
Increasing use of anti-VEGF monoclonal antibody- very effective

Diabetes is associated with increased mortality & morbidity due to its microvascular & macrovascular complications

COMPLICATIONS





Chronic Diabetic Complications

Microvascular (small vessel disease)

- Retinopathy
- Nephropathy (glomerulosclerosis):
microalbuminuria; albuminuria (proteinuria);
raised serum creatinine; end-stage renal failure
- Neuropathy: peripheral; autonomic;
acute mononeuropathy; diabetic amyotrophy

Macrovascular (large vessel disease)

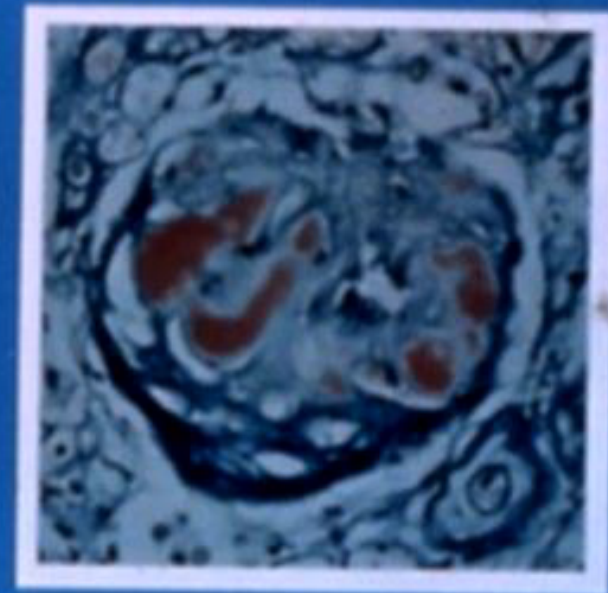
Stroke; coronary and peripheral artery diseases

Diabetic Nephropathy

CLINICAL FEATURES

1. Proteinuria
2. Nephrotic oedema
3. Hypertension
4. Renal failure

HISTOLOGY



Chronic Diabetic Neuropathy



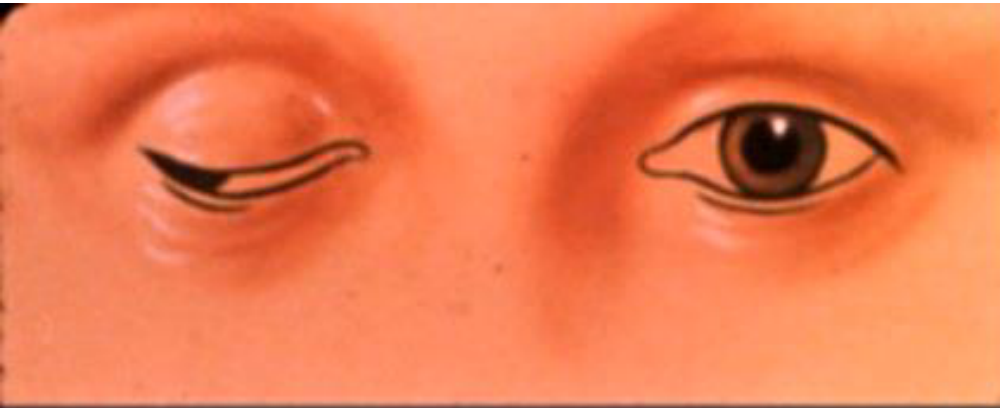
Sensory disturbance

Motor disturbance

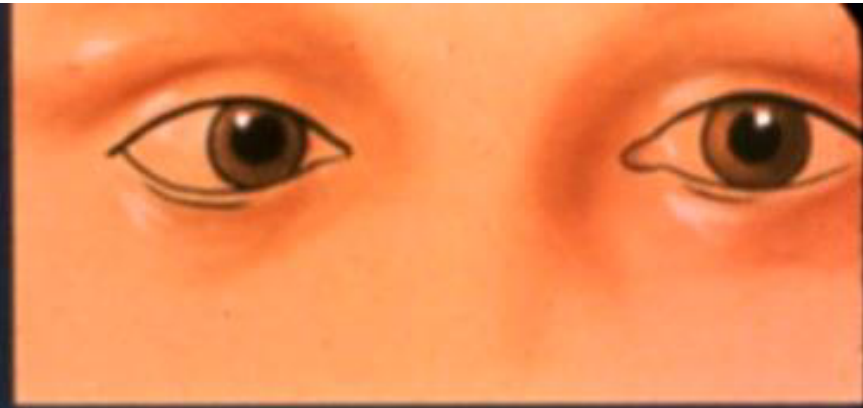
Autonomic disturbance

30% patients

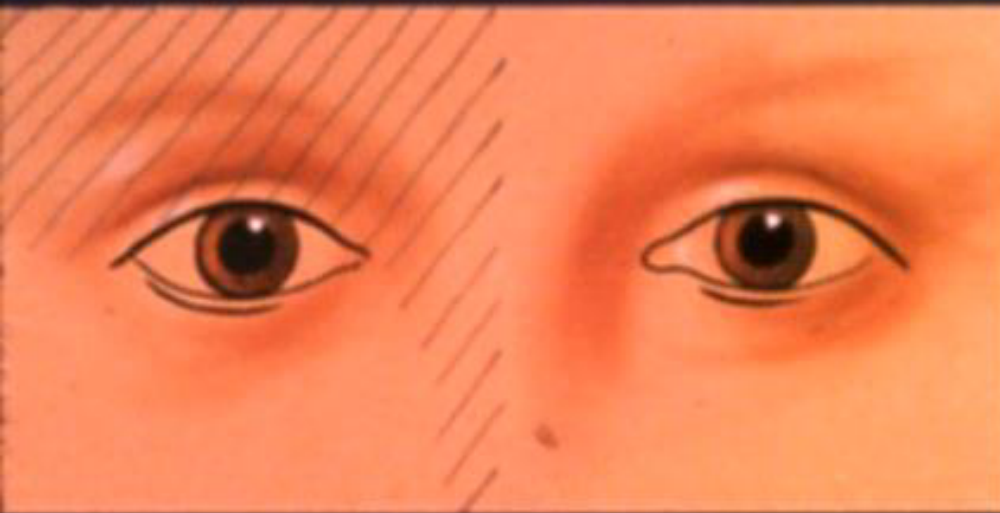
Diabetic Mononeuropathy: Acute



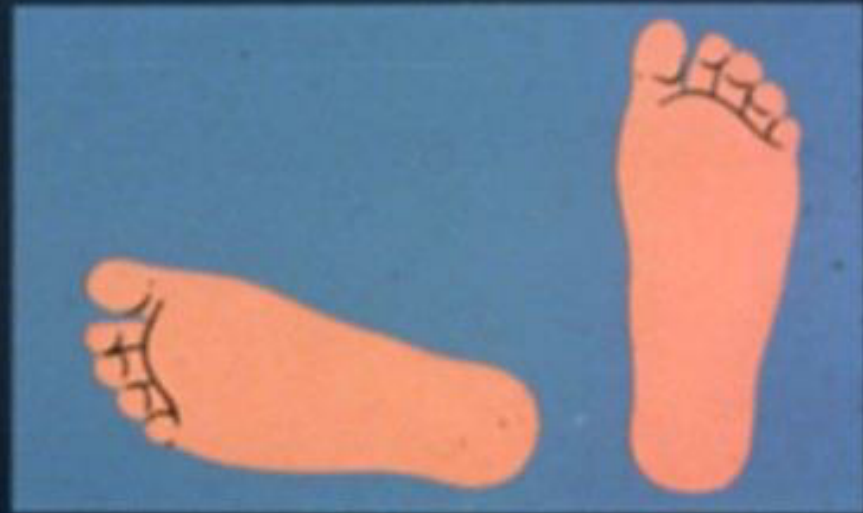
Ptosis & Divergent squint



External Rectus Palsy

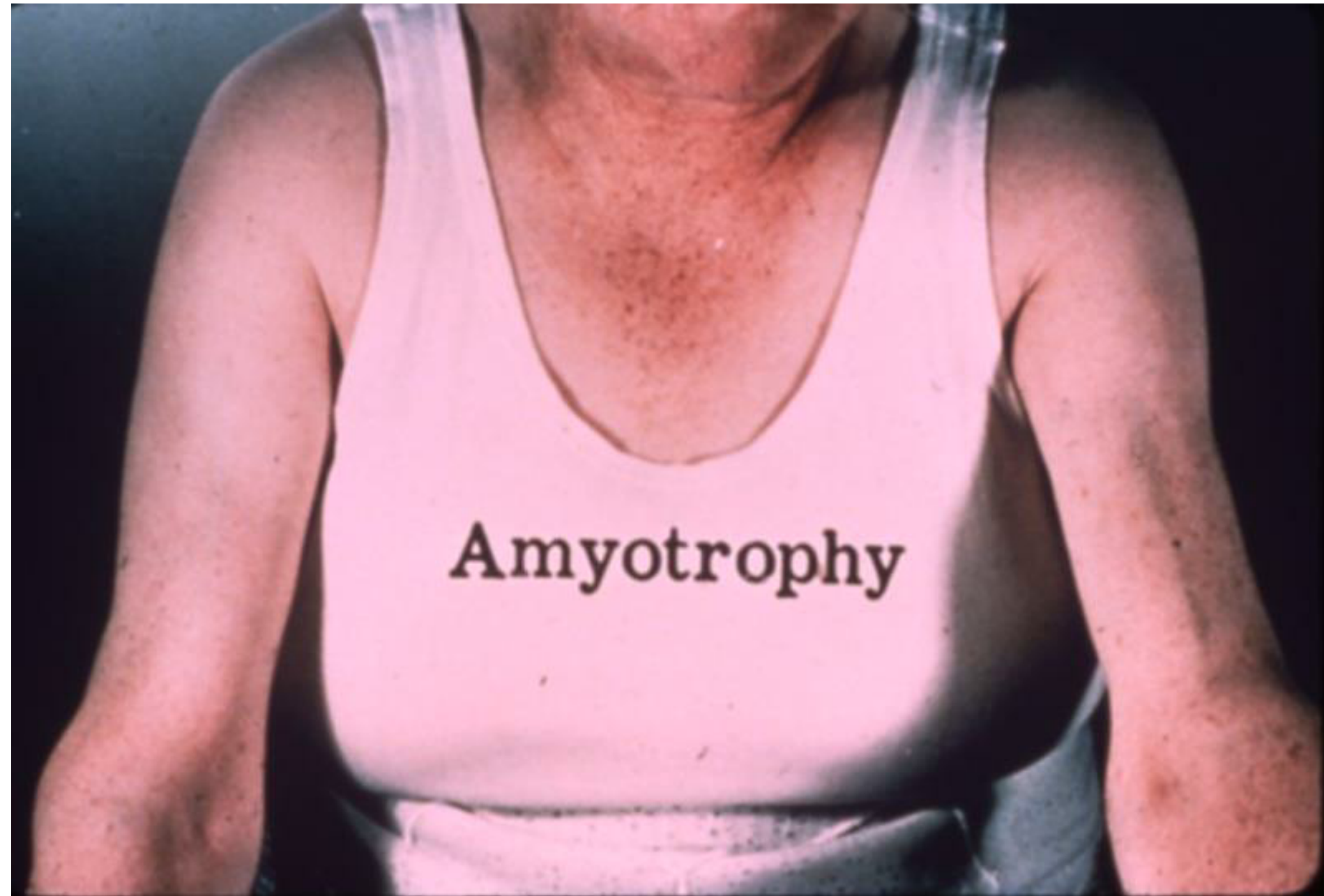


Upper facial & Eye pain

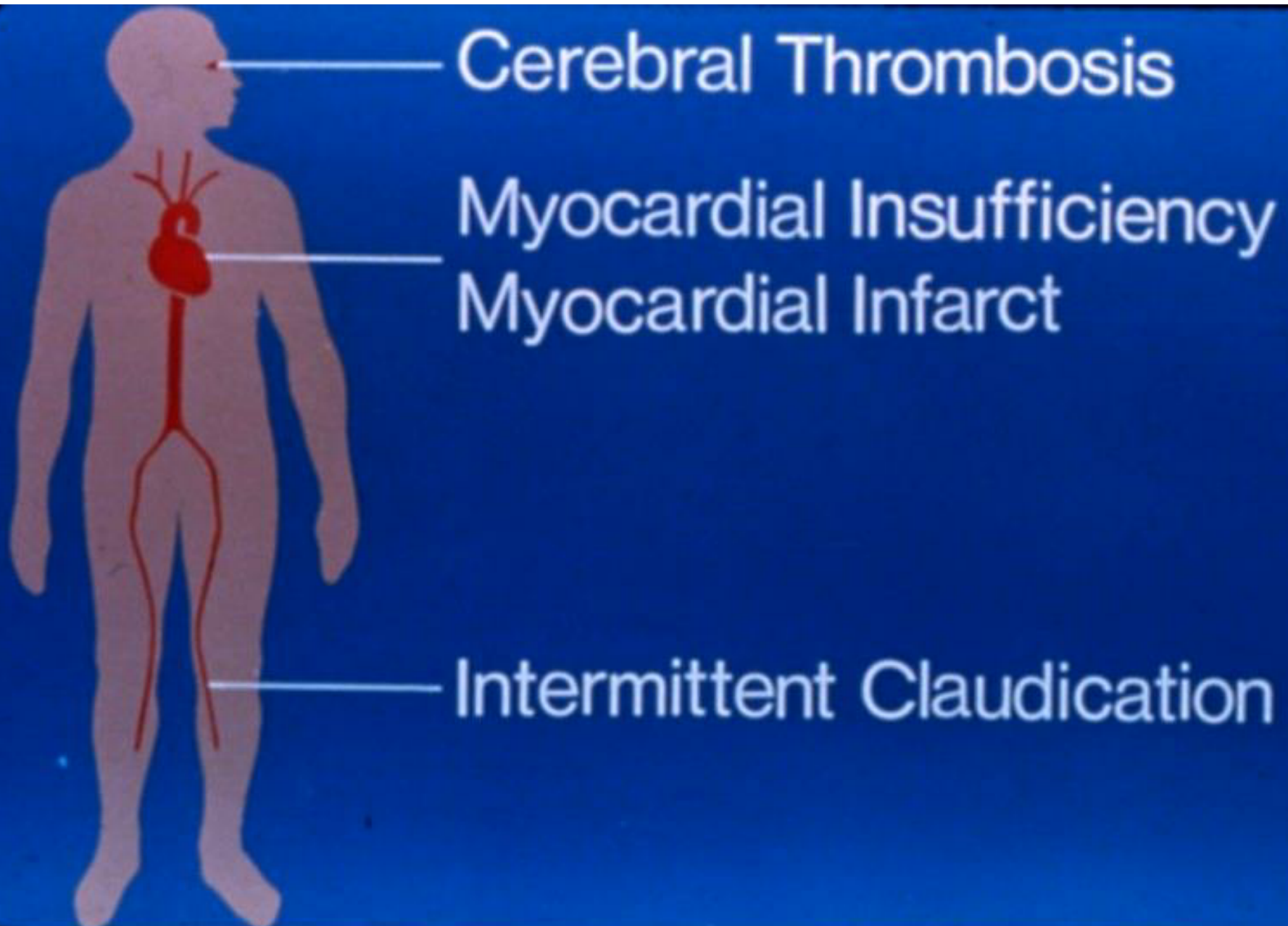


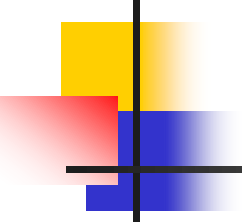
Foot drop

Diabetic amyotrophy: acute/subacute



Macrovascular Diabetic Complications

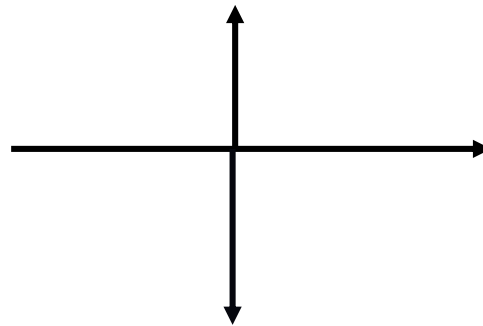




Pathogenesis of Chronic Diabetic Complications

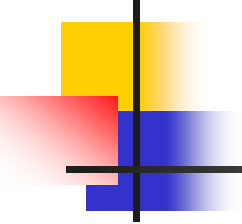
Genetic predisposition

Prolonged
hyperglycaemia



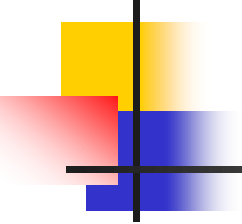
Chronic diabetic
complications

Accelerating factors
(↑ BP, etc)



Treatment of chronic complications - principles

1. Prevention - good glycaemic control
2. Insipient (subclinical) stage - reversible
e.g. microalbuminuria
 - improve glycaemic control
 - treat other risk factors - BP, smoking
 - pharmacological intervention:
 - Angiotensin converting enzyme inhibitor (ACEI)
 - Angiotensin II receptor blocker (ARB)
 - SGLT2 inhibitors/GLP-1 receptor agonists
 - Finerenone (non-steroidal aldosterone receptor antagonist)



Treatment of chronic complications - principles

Overt complications

Albuminuria, moderately severe retinopathy; clinical neuropathy

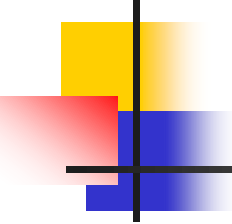
Progression ↓ by:

General: ↑glycaemic control; risk factor management
(hypertension, hyperlipidaemia, smoking)

Specific: e.g. nephropathy – renin-angiotensin blockade (ACE inhibitors; angiotensin receptor blocker); SGLT2 inhibitors; GLP1 receptor agonists, finerenone.

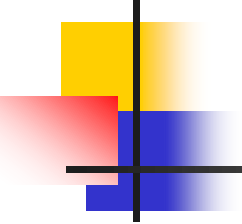
Symptomatic - dialysis / transplantation; pain relief

Prevention of drastic consequences - laser therapy; foot care



How to achieve glycaemic control

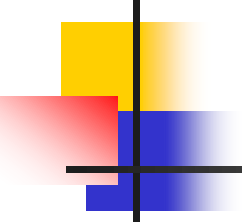
- Lifestyle measures: diet + exercise; optimise body weight
- Lifestyle measures + oral hypoglycaemic agents
- Lifestyle measures + insulin



Dietary management: integral part of therapy for all patients with diabetes

General guidelines:

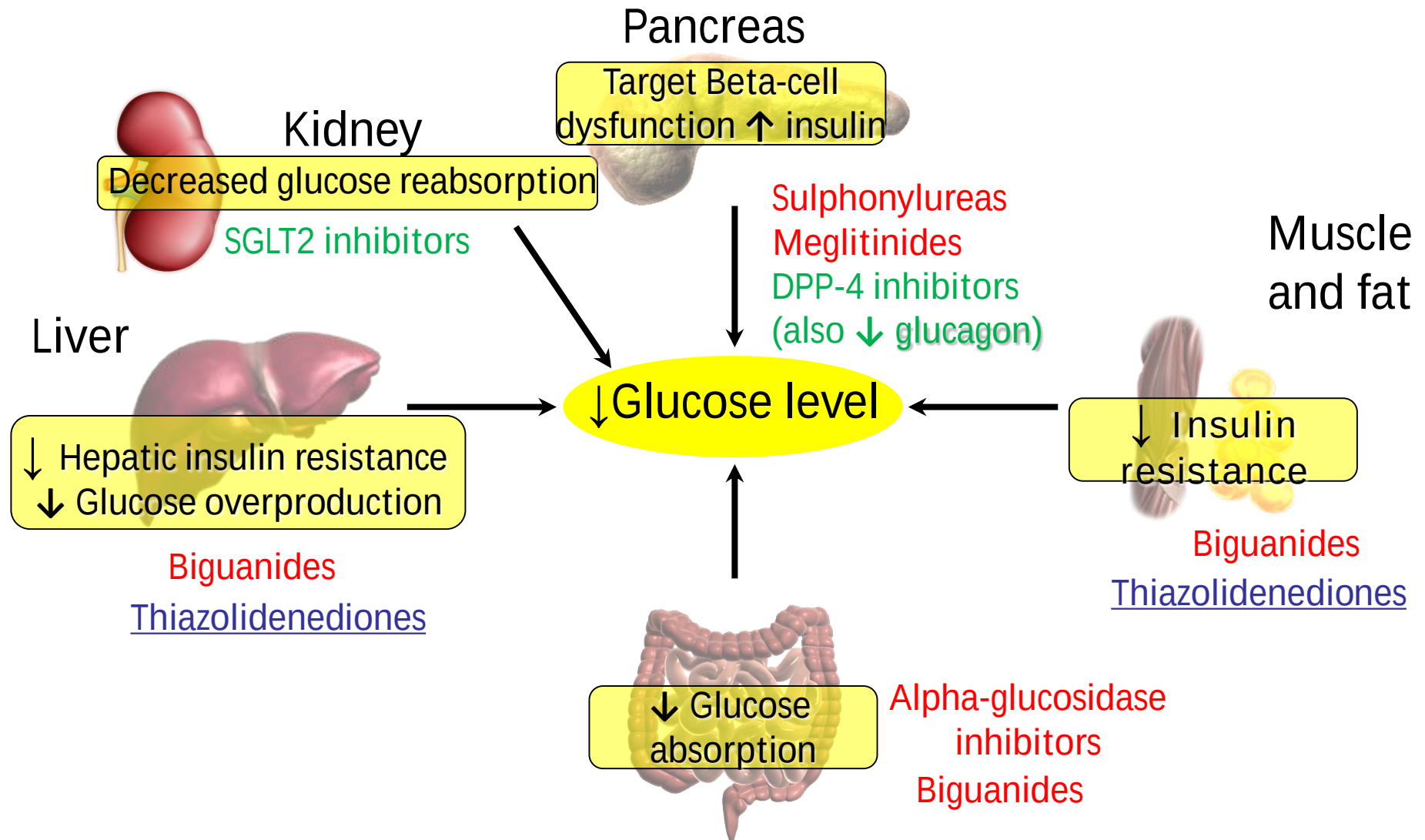
- 30 Kcal/Kg ideal body wt. per day, with adjustments depending on life style and body weight
- Aim at achieving normal body weight.
- Composition:
 - 40-50% Carbohydrates
 - 30% Fat (<7% saturated fat)
 - 20-30% Protein
 - Fibre intake 20-35 gm per day



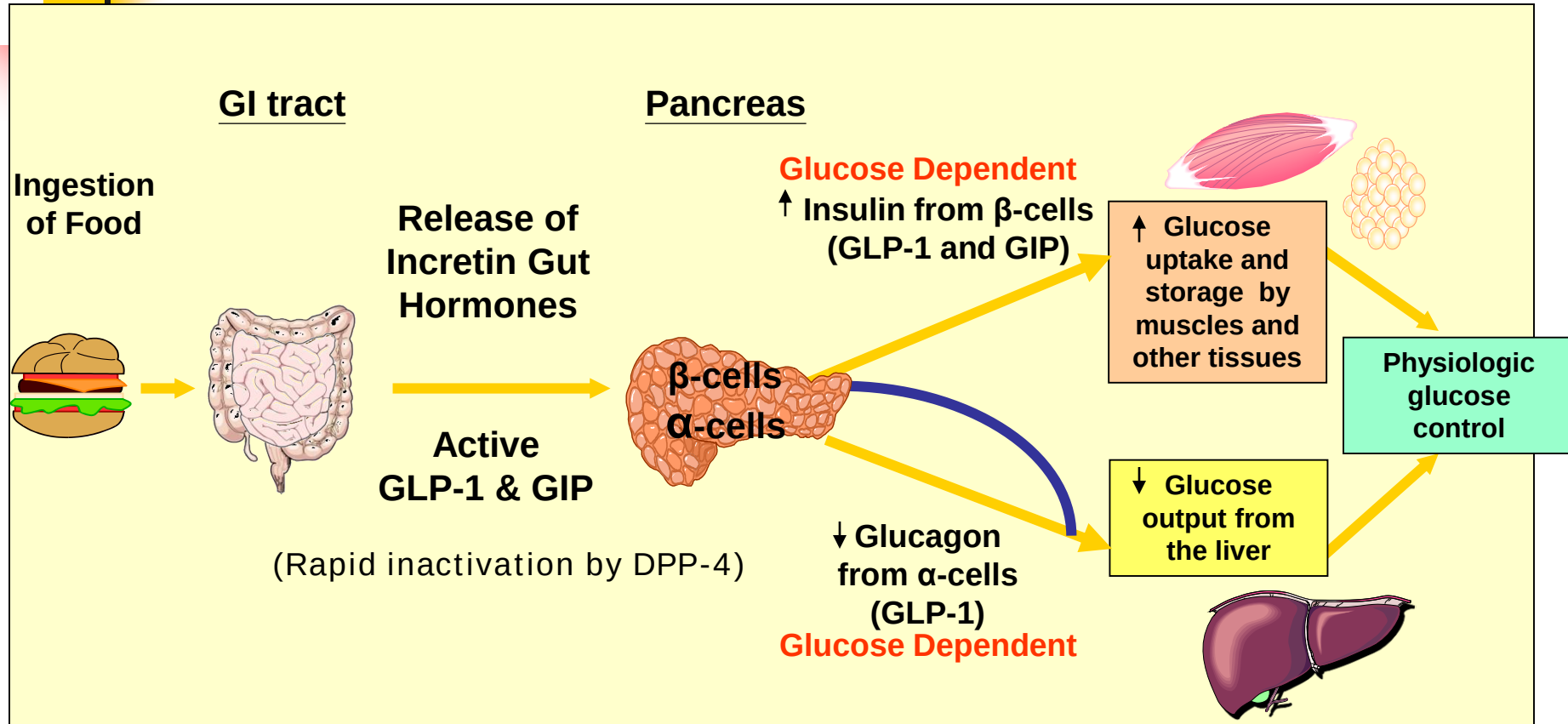
Dietary advice

1. Personalised: according to individual preference & culture
2. Consistency of meal timings & quantity - especially if requiring insulin
3. Healthy foods consistent with prevailing population-wide dietary recommendations
4. Emphasize on high fibre foods (vegetables, fruits, wholegrains & legumes), low-fat dairy products, fresh fish
5. Minimise high energy foods (high saturated fats, sweet desserts and snacks)
6. Hypocaloric diet for obese type 2 patients

Major Targeted Sites of Oral Drug Classes



Incretins Physiologically Regulate Insulin and Glucagon in a Glucose-dependent Manner



The “Incretin Effect” is diminished in patients with T2DM:
Effect of GIP on β -cells is diminished

Adapted from Kieffer TJ, Habener JF. *Endocrine Reviews*. 1999;20:876–913.; Ahrén B. *Curr Diabetes Report*. 2003;2:365-372.; Drucker DJ. *Diabetes Care*. 2003;26:2929-2940.; Holst JJ. *Diabetes Metab Res Rev*. 2002;18:430-441.; Drucker DJ. *Expert Opin Investig Drugs*. 2003;12:87-100.; Ahrén B et al. *Horm Metab Res*. 2004;36:867-876.

Incretin Based Anti-diabetic therapy

No hypoglycaemia if monotherapy

GLP-1 receptor agonists: Stimulate GLP-1 receptor

- Injectable peptides (except oral semaglutide) resistant to DPP-4
- ↑ post-prandial insulin release & glucagon suppression
- Delays gastric emptying; reduces appetite: ↓ **body weight**
- Twice daily: exenatide - Daily: lixisenatide, liraglutide
- Weekly: exenatide LAR; dulaglutide; semaglutide (SC/oral)
- **Cardio-renal benefits:** liraglutide; dulaglutide; semaglutide
- Can be used with insulin and other ADMs except DPP4i

GLP-1/GIP dual agonists: eg. Tirzepatide – even higher efficacy

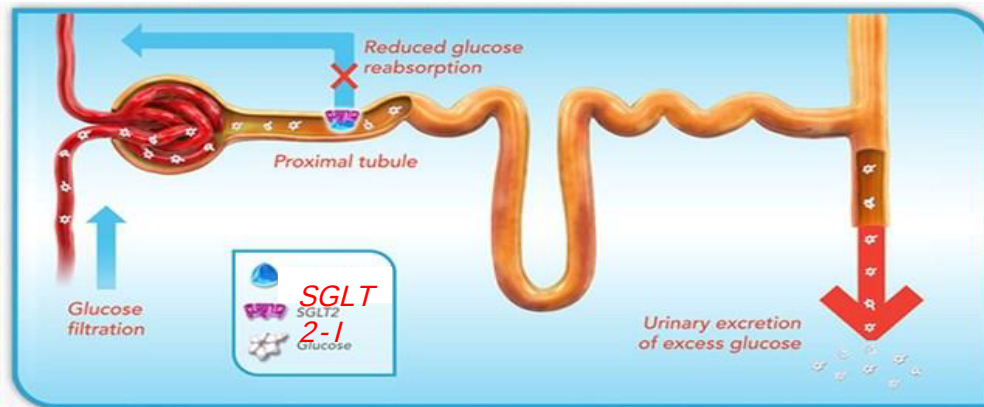
DPP-4 inhibitors (DPP-4i): **weight neutral**

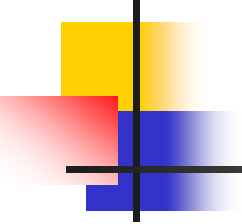
- Oral agents; increase endogenous active incretins by inhibiting DPP-4: Sitagliptin, Vildagliptin, Saxagliptin, Linagliptin; Alogliptin; available also as fix-dose combinations with metformin, pioglitazone, SGLT2 inhibitor. Can be used with other anti-diabetic medications

Treatment considerations: Renal dose for DPP-4 inhibitors (except linagliptin); GLP-1 RA: if eGFR < 30 ml, not to use exenatide

Selective sodium-glucose co-transporter 2 inhibitors (SGLT2i)

- ↓ hyperglycaemia by increasing urine glucose loss
- ↓ weight, fat mass and BP; no hypoglycaemia if monotherapy
- Cardio-renal protection; reduces heart failure even in patients without diabetes (via non-glycaemic effects such as interaction with renal Na/H exchanger, improved renal function, etc)
- Can be used with other anti-diabetic drugs
- Approved: Dapagliflozin; Canagliflozin; Empagliflozin; Ertagliflozin





Side-effects of oral diabetic drugs

Metformin: Dyspepsia, diarrhoea

Lactic acidosis - rare except in presence of renal/ hepatic/
cardiac failure/alcoholism/ myocardial infarction/ septicaemia

B12 deficiency may occur with long-term high doses

Not to be used if $\text{eGFR} < 30 \text{ mL/min/1.73m}^2$

Withhold before & for 48h after contrast imaging if $\text{eGFR} < 60 \text{ mL/min}$,
liver disease, heart failure, alcoholism or intra-arterial contrast

Sulphonylurea & meglitinides (eg repaglinide): Weight gain +;
hypoglycaemia; insulin preferred if renal insufficiency

α -glucosidase inhibitor (Acarbose): Flatulence; diarrhoea;
glucose required for relief of hypoglycaemia



Side-effects of oral diabetic drugs (cont)

Thiazolidinediones (TZD; pioglitazone):

Fluid retention: contra-indicated in heart failure;

Weight gain ++; increase fractures; anaemia; ?? increase Ca bladder

DPP-4 inhibitors:

Pancreatitis (very uncommon); rarely joint pain

SGLT2 inhibitors:

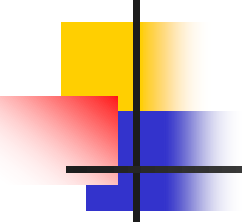
Genital infection (fungal); lower urinary tract infection;

Osmotic diuresis and postural hypotension;

Rare cases of diabetic ketoacidosis (risk factors: longstanding diabetes; low CHO intake; prolonged fasting; dehydration; alcoholism). Stop 3 days before surgery or endoscopy.

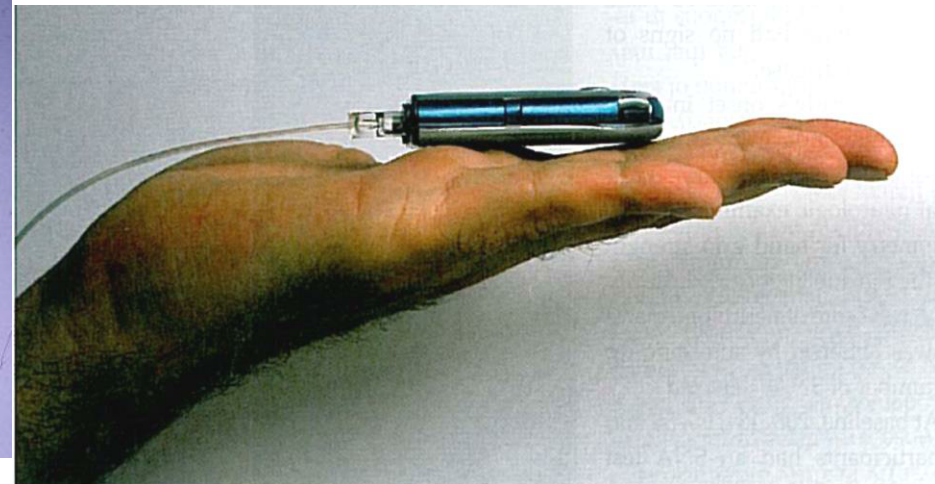
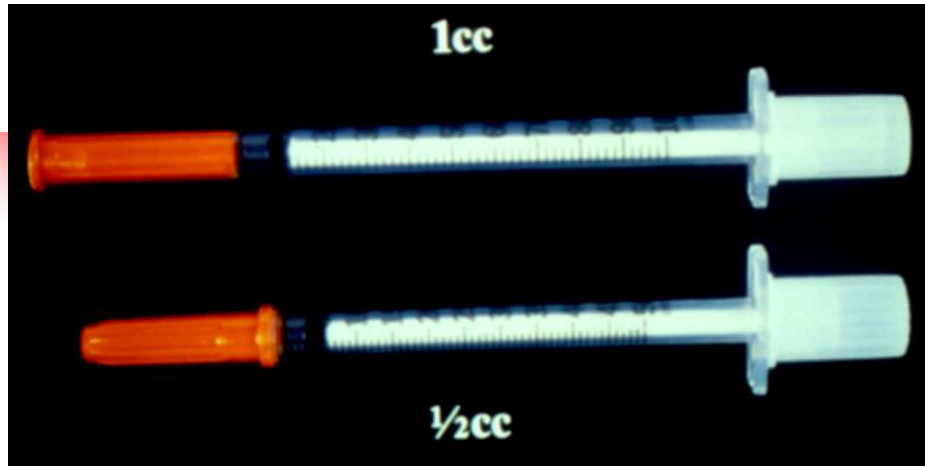
Dapagliflozin: avoid if active Ca bladder

Canagliflozin: may cause low bone mass/increased fracture risk



Insulin

- Subcutaneous injections 1-4 times per day; needle & syringe or insulin pen: for maintenance treatment
- Intravenous infusion or intramuscular injection:
for acute or temporary treatment of hyperglycaemia
- Continuous subcutaneous insulin infusion (insulin pump)
- Inhalation insulin: high cost; low patient acceptance;
??↑ lung cancer; only one in clinical use (not in Hong Kong)
- Source : Recombinant human (HM) or insulin analogs
Bovine/porcine insulin (replaced by HM insulin)



A miniature insulin pump.

Insulin Pump + CGMS

REAL-Time Insulin Pump and
Continuous Glucose Monitoring
System

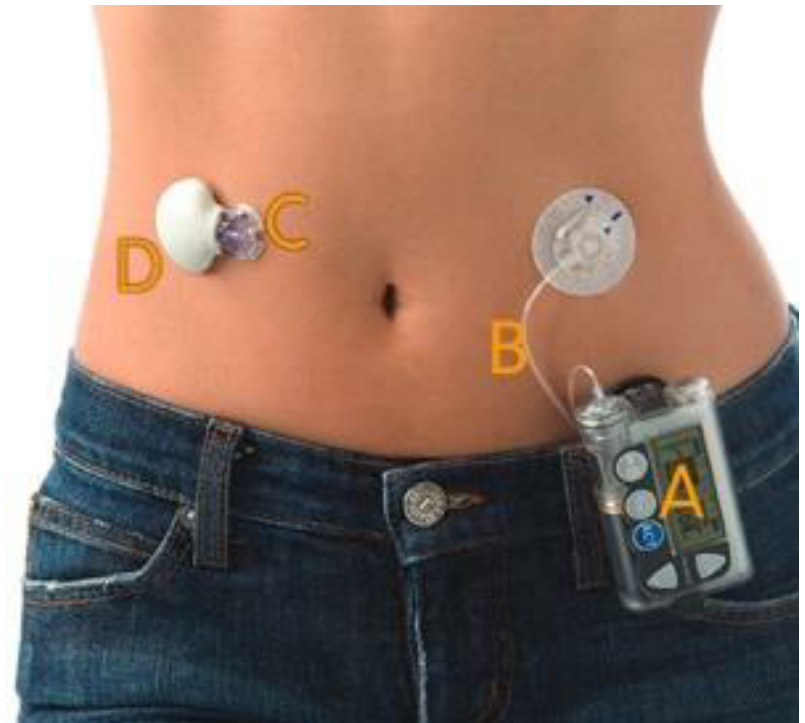
A = Insulin Pump

B = Cannula

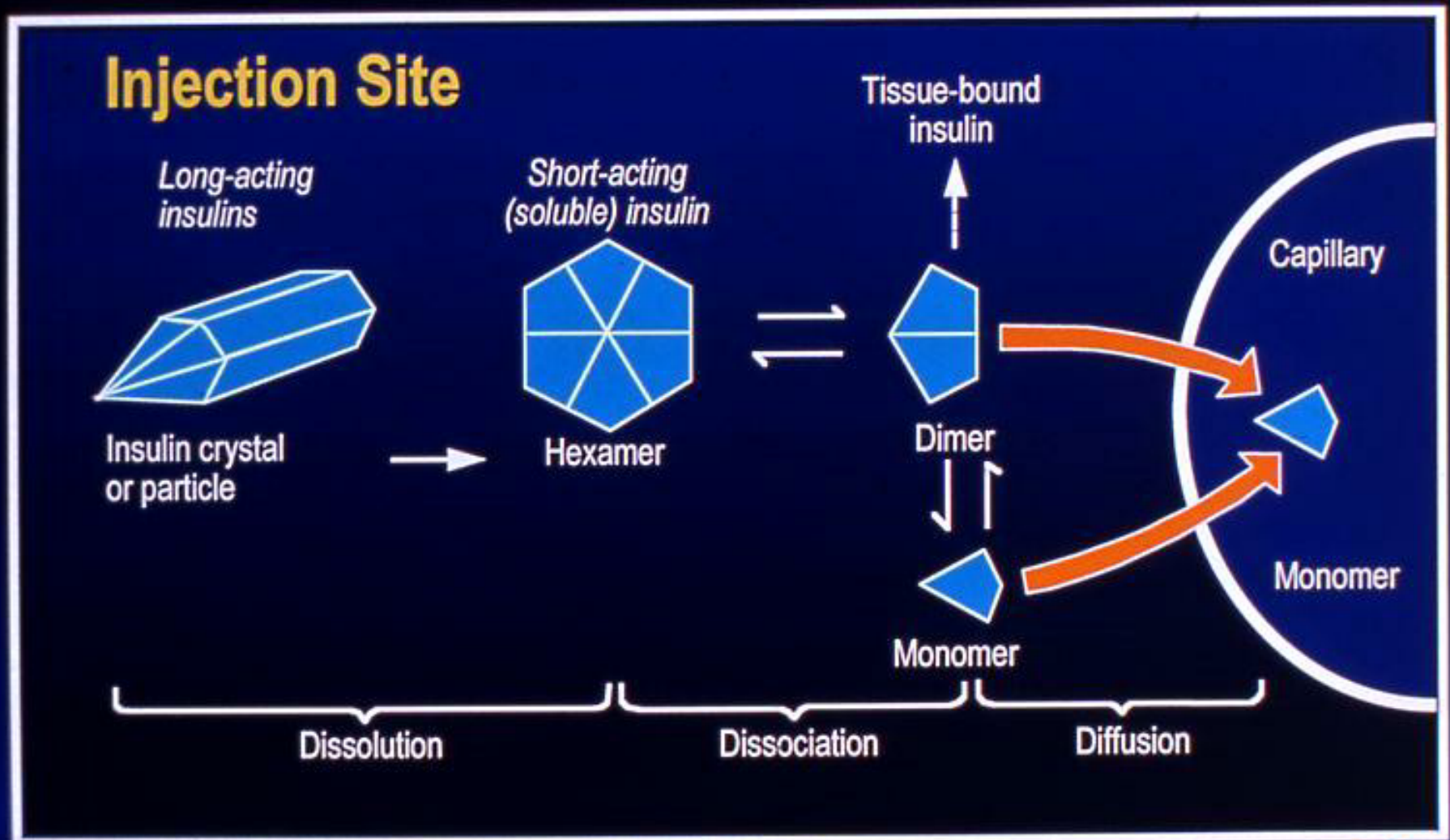
C = Sensor

D = Transmitter

Artificial pancreas = close-loop
system with direct A to C
inter-talk; patient input intake
via smart phone (FDA approved)



Self-association of insulin after subcutaneous injection

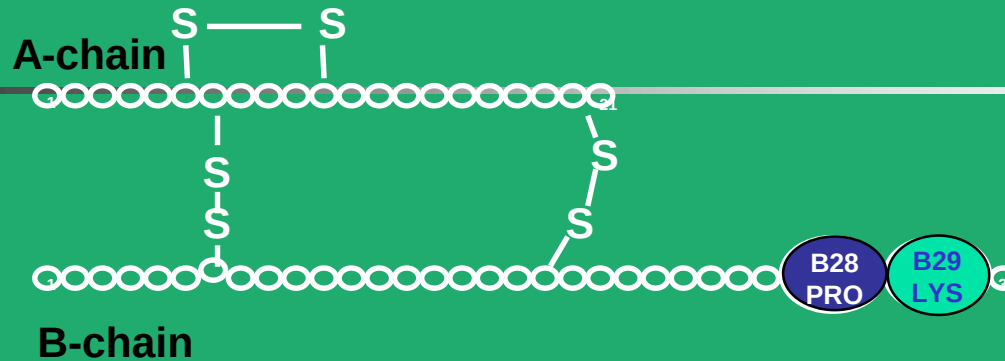


Textbook of Diabetes, Vol. 1; Oxford: Blackwell Scientific Publications, 1991; 375.

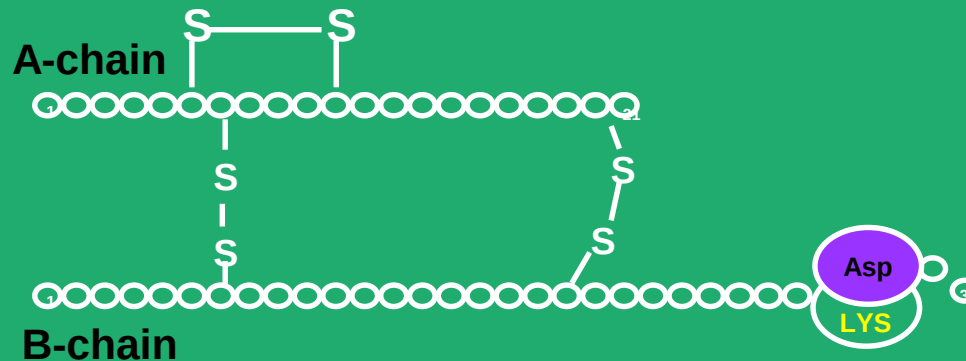
Lilly

18.6.20

Short-acting Insulin Analogues (Modified Insulin Molecules)



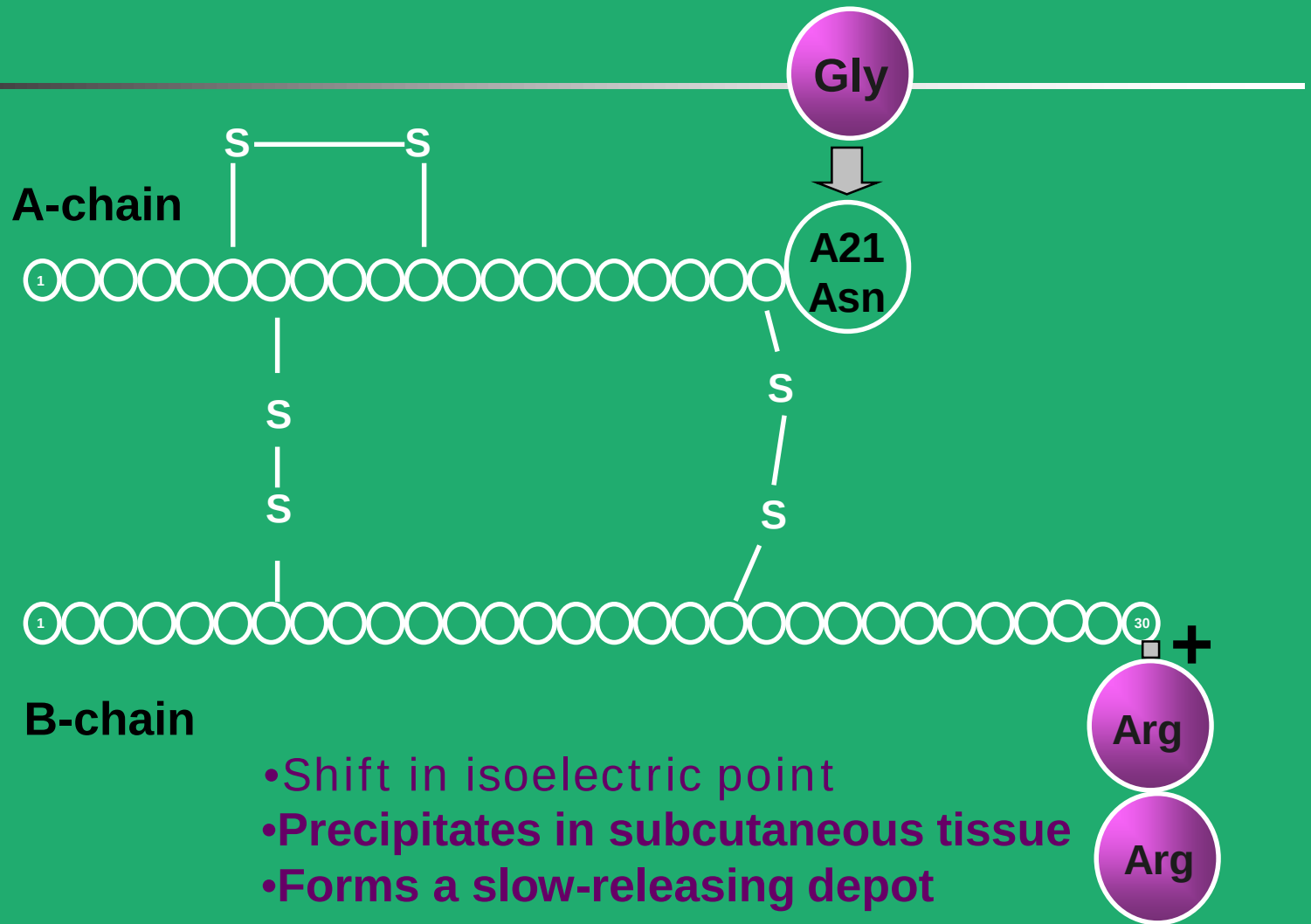
Insulin Lispro
(Humalog)

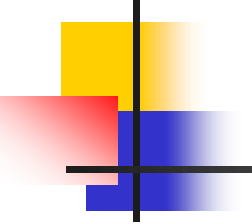


Insulin Aspart
(Novorapid)

Short-acting analogues: less self-association than human insulin after SC injection
More physiological profile: faster onset; shorter duration; less premeal hypoglycaemia

Long-acting insulin analogues – Glargine (Lantus)

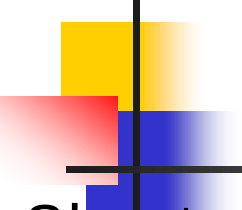




Long-acting Insulin Analogues:

Delayed onset & sustained action with steady, predictable plasma level:

- Glargine: precipitates in subcutaneous tissue; forms a slow-releasing depot
- Less nocturnal and symptomatic hypoglycaemia compared to insulins with protamine as retardant (protaphane; Humulin N)



Insulin with different duration of action

Short-acting: Regular human insulin – Actrapid, Humulin R

Rapid-acting analogs - Insulin Aspart; Insulin Lispro; Insulin Glulisine

Ultra-Rapid Acting: lispro-aabc; fast action insulin aspart (Fiasp)

Intermediate-acting (protamine as retardant):

Protaphane; Humulin N

Premixed human: Mixtard 30 (protaphane/actrapid 70/30); Novo-Mix 30

Humulin 70/30; Humalog-Mix (25/75; 50/50);

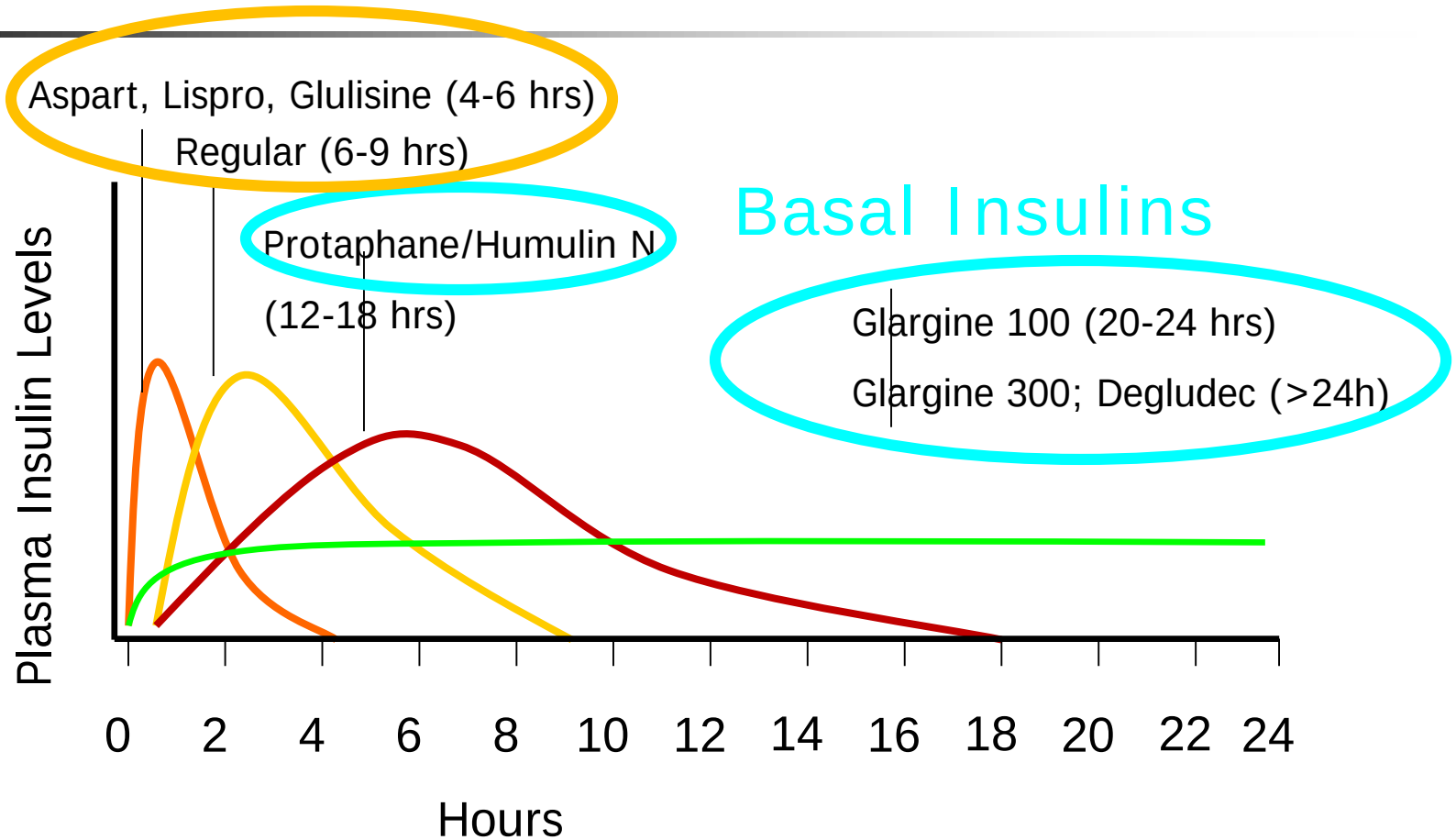
Long –acting analogs: Glargine (Lantus)

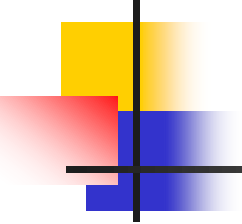
Ultra-long acting analogs: Glargine 300 u/ml ($T_{1/2}$ =19 hours vs 12.1 hours for Lantus); Insulin Degludec ($T_{1/2}$ =25.4 hours)

Weekly insulin: Insulin Icodec (Awiqli); Efsitora alfa

Human Insulins and Analogues

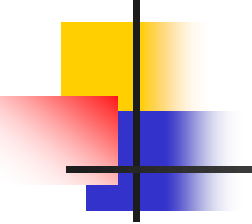
Prandial Insulins





Insulin requiring patients with diabetes

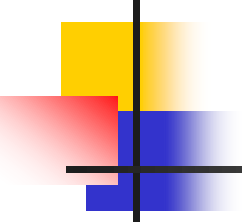
- Marked recent weight loss
- Marked ketosis
- Underweight; clinically ill (dehydrated, infection, infarction)
- Severe hyperglycaemia at diagnosis: HbA1c >10%; fasting glucose >16.7 mmol/L (300 mg/dl)
- Pregnancy
- Failed oral therapy (GLP-1RA as preferred alternative if HbA1c<11%, especially if obese, presence of cardiovascular disease [CVD] or chronic kidney disease [CKD])



Management of hyperglycaemia in type 2 diabetes (ADA)

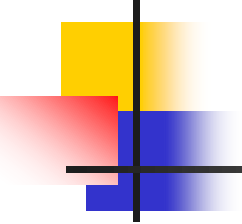
Healthy lifestyle + medication choice according to treatment goals

- I) Reduction of cardiovascular (CV) and renal risk if high risk or established CVD/CKD: SGLT2i (esp if heart failure+), GLP1-RA or both preferred; avoid TZD and saxagliptin if heart failure
- II) Achievement of weight and glycaemic goals
 - Metformin: glycaemic and weight effects superior to DPP4i & sulphonylurea
 - Increasing emphasis on weight as treatment goal: for glycaemia, CVD and steatotic liver disease reduction: SGLT2i/GLP-1RA/tirzepatide
 - Compelling need to minimize hypoglycaemia: assess need/dosage of drugs with higher risk (sulphonylureas, meglitinides, and insulin)
 - Initial combination therapy should be considered if presenting with A1C levels 1.5–2.0% above individualized goal.
 - Consider accessibility and cost of medications.



Comprehensive Management of Diabetes

- Patient education and support– healthy lifestyle, self-management; team approach
- Treatment of diabetes - diet, oral drugs, GLP1 agonists and co-agonists; insulin
- Treatment of associated coronary risk factors - hypertension, ↑lipids, smoking, physical inactivity, obesity
- Individualised treatment targets/choices
- Regular assessment for complications



JC Whole Class Session

TELL US
WHAT
YOU THINK

THANK
YOU!



Look for today's teacher
and the others you may have
(HKU Portal login required)
missed.